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Comparison Between Two Control Algorithms for EMG-Based Navigation

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Abstract – The purpose of this study was to develop and compare two control algorithms that could potentially utilize two channels of surface electromyogram (EMG) to navigate electromechanical devices such as a powered wheelchair in all directions, i.e. forward (F), backward (B), right (R), and left (L). The first algorithm uses a non-timed muscle twitch to change between six modes (F, B, FR, FL, BR and BL), while the second algorithm uses a quick-timed twitch to switch between two modes (F and B). The second portion of this study consisted of acquiring EMG signals from the muscles of the forearm through a simple instrumentation circuit. The study was focused on evaluating the effectiveness of the controlling schemes. While the second algorithm offered faster manipulations, the first algorithm was more reliable and resulted in fewer collisions.

I. INTRODUCTION

Electromyogram (EMG) signals can be used as an application to assist people with paralysis in controlling powered devices [1, 2]. This project is a continuation of a previous project design to use EMG signals for controlling an electromechanical device [3]. The present study is focused on developing and comparing two control algorithms in terms of their effectiveness for EMG-based navigation of a powered device. A remote-control model car is used in the experiment. The two algorithms were based on a single switch or quick twitch respectively, with potential integration with previous studies dealing with rising edge control, pulse width control, and falling edge control.

II. METHODS

A. Development of the Algorithms

The software is developed in a C environment and consists of two different algorithms. The hardware involves an inexpensive PIC18F452 microcontroller, as well as two simple on/off switches. In addition to this hardware, there is a set of six light-emitting diodes (LED's) to indicate the mode within each separate algorithm. In consideration of safety, the algorithms have been programmed so that if neither of the switches are pressed, the motorized device will not move. It will only move if the correct switch is pressed, and will stop as soon as the electrical impulse ceases.

The first algorithm has six modes, each corresponding to an LED light. The first mode is a simple forward mode and the next is the backward mode. The modes follow sequentially in

this order: forwards and to the left, forwards and to the right, backwards and to the left, and lastly backwards and to the right. In order to switch between the modes, user must hit the left switch. This advances the modes in a forward order. It is also possible to advance the modes in a reverse order, to decrease the hassle of scrolling all the way back through the list. In order to do this, the user must hold the left switch down and press the right switch until the light panel indicates the desired mode. After the mode is selected, the user only has to press the right switch, which sends a signal to the motorized device to move.

The second algorithm only has two modes. The first mode is a forward mode. Within this mode, the user must press the left button to move forwards and to the left, the right button to go forwards and to the right, and both buttons simultaneously to go forwards in a straight path. A quick twitch on the right switch changes the mode, from forwards mode to backwards mode, and back again. In the backwards mode the controls are the same as for the forwards, but the car moves backwards.

B. EMG Instrumentation

The control sequence that will drive the motorized device will require an input from the user. The input will be EMG signals collected from the forearm of the user. As shown in Fig. 1, the circuit that receives the EMG signals was built by use of an instrumentation amplifier (AD620, Analog Devices, Norwood, MA). For the signals to be used by the PIC processor, further amplification was needed, and was done by a second amplification stage made from a LM324N operational amplifier. The amplifiers were biased to have a gain of 50, combined. A potentiometer was used in the second stage to bring the signal rails from -9 to 9 volt range to the 0 to 5 volt range, required by the PIC processor. The analog signals coming from the amplifier are then digitized by the PIC18F452 microcontroller to be used by the control scheme.

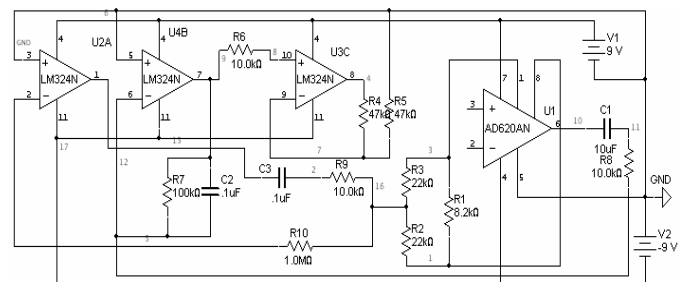


Figure 1. Schematic diagram of EMG instrumentation circuit.

The circuit required three electrodes, two for input signals and one for an active ground. The electrodes used were made of solid copper disks.

C. Testing

To evaluate the two algorithms, the radio frequency control circuitry from a model care was wired to the processor. The model car was navigated through an obstacle course by using the two switches. To obtain evaluation data, three individuals tested the two control schemes on an obstacle course four separate times. As shown in Fig. 2, the course was split into two sections. The “figure 8” tested driving that required less accuracy and allowed for higher speeds. The “parallel parking” section of the course tested more precise, and consequently slower, movement. Each participant was allowed to practice with the algorithm until they felt reasonably comfortable driving the device. At that point, the time it took each user to run through each segment of the course was recorded. The number of times the user “crashed,” by causing the device to impact a wall or obstacle, was also recorded. An analysis of the time it took to complete the course and the number of “crashes” allowed for an objective way to compare the two algorithms. In addition to the quantitative data, the observations of the participants while they were using the algorithms were considered when evaluating the success of each control scheme.

III. RESULTS

The average time it took to complete the run was slightly faster when using Algorithm 2, quick twitch, as opposed to Algorithm 1, single switch (Table 1). The difference was about 12%. This held true for all but two of the individual “Figure 8” tests. However the total number of crashes was higher for Algorithm 2 when compared with Algorithm 1, for both the “Figure 8” and the “parallel parking.” Algorithm 2 averaged 1.5 additional crashes (Table 2). The users reported that Algorithm 1 was more problematic to use due to the dual affects of the need to stop and switch modes, as well as the time required pressing the left switch to scroll through the modes. Users unanimously agreed that Algorithm 2, with the two modes and more traditional control scheme, was a more natural method of driving. Observation attests to this as participants were much more likely to travel at higher speeds while using Algorithm 2 than Algorithm 1. However participants at times had difficulty facilitating the “quick twitch” required on the right switch to change the mode. They reported that it sometimes required multiple “twitches” to change the mode. At one instance the switch got stuck and the

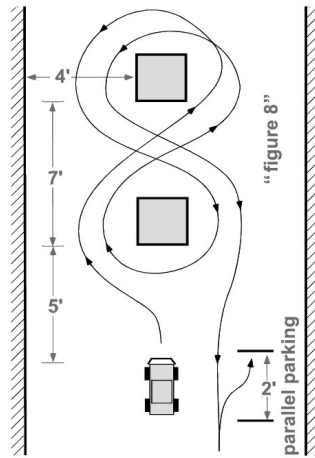


Figure 2. Obstacle course.

Average Time Per Run (sec)		
	Algorithm 1	Algorithm 2
Figure 8 Time	53	44
Parallel Parking Time	46	43
Total Time	99	87

Table 1. Comparison of average time per run

Average Number of Crashes Per Run		
	Algorithm 1	Algorithm 2
Figure 8 Crashes	0.5	1.4
Parallel Parking Crashes	0.1	0.6
Total Crashes	0.6	2

Table 2. Comparison of average number of crashes

car continued moving even after the switch was let up. It required another twitch to fix the situation.

IV. DISCUSSION

Two control algorithms for navigation using two switching signals were evaluated in this study. Algorithm 1 uses one switch to choose among six directions and the other switch to drive. Algorithm 2 uses a quick twitch to choose between two directions and both switches to navigate. Although users expressed discontent with Algorithm 1 in the learning phase, they felt comfortable after multiple practice runs through the obstacle course. Users were able to navigate with a much higher degree of ease and at faster speeds. The average speed difference between the two algorithms was not large enough to afford Algorithm 2 a large advantage over Algorithm 1. Also, the number of crashes was lower for Algorithm 1 than Algorithm 2. One of the most important factors in this comparison was the participants’ repeated difficulty with facilitating the “quick twitch” in Algorithm 2, which demanded a high degree of dexterity. If users were having trouble with that aspect of the program, it is expected that further difficulties will arise when the same feat is attempted using EMG signals. Previous research has shown that short muscle twitches combined with single switching methods can help users to navigate slowly to the proximity of objects [2]. Thus, Algorithm 2 may still be a viable control scheme for persons with sufficient dexterity, but Algorithm 1 appears to be a more reliable approach in general.

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